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Economic Report and Production Analysis for Egypt

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0. Executive Summary

In this report, we aim to uncover the economic path Egypt went through during the period from 2009 to 2014. Specifically, we dive into the topic through a macroeconomic overview of Egypt from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We find that during the period of our interest, Egypt demonstrated a moderate level of economic growth with an annual GDP growth rate of 10.8%, concurrent with a slight decrease in economic complexity as compared with other major emerging economies.

The level of economic progress during the period, we believe, can be partially explained through analysis of the economic and product complexity structure of Egypt on the industry group level. The product classification system we adopt allow us to measure the actual development for various industry groups and to observe that Egypt invested heavily in developing animal, vegetable, mineral and stone industries during the period, while its emerging peers and developed economies around the globe have adopted much more complex and progressive paths.

Based on our analysis of the opportunity gain and a sensitivity analysis of major industry groups, Egypt could further its gains by high-tech, machinery, and daily instrument. We further note that Egypt has a slightly matched development strategy that indicates a moderate level of structural optimality on the complexity level, having an adjusted R-square of around 19.2%. Since a significant positive correlation exists between five-year GDP growth and the structural optimality index of an economy as we define and calculate, Egypt indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale, and has the potential to further its economic gains through industrial structure optimization as suggested in the paper.

1. Report Overview

In this report, we aim to uncover the economic path Egypt went through during the period from 2009 to 2014. Specifically, we dive into the topic through a macroeconomic overview of Egypt from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We then analyze the industrial framework and changes of Egypt during the period based on quantitative measures such as Product Complexity Index (PCI) and Product Family Complexity Index (PfCI), followed by a detailed analysis of how such changes have matched with development strategies as quantitatively measured by opportunity gains on an industry-level. We conclude our report with macro-level results and predictions of Egypt, as deduced from regression models we have established on a global scale, which takes into account countries' stances in their product space.

2. Macroeconomic Results

2.1 GDP Ranking and Changes

As the largest economy in Africa by GDP, Egypt has a GDP of 188.98 billion USD in 2009 and 305.53 billion USD in 2014, exhibiting a promising level of economic output. Figure 1 illustrates the time-series GDP results for top African countries from 1996-2016, and Egyptranks among its counterparts as a major African economy.In particular, the period from 2009 to 2014 seems to be a rather stable period for Egypt in terms of GDP growth as compared with its emerging peers. During the five years, Egypt exhibited an annual GDP growth of 10.08%, comparable with the annual growth rate of the other five leading African economies, which have a geometric average growth rate of 10.89% and an arithmetic average growth rate of 12.25%. Also, Egypt's GDP ranking in a global scale shifted from the 43rd to the 38th place from 2009 to 2014, confirming a moderate level of economic progress during the period. The economic growth during the period, we believe, can be reflected through changes in ECI and PCI, and can be further explained by a production strategies adopted by Egypt during the period, which we will expand at the later stage of the report.

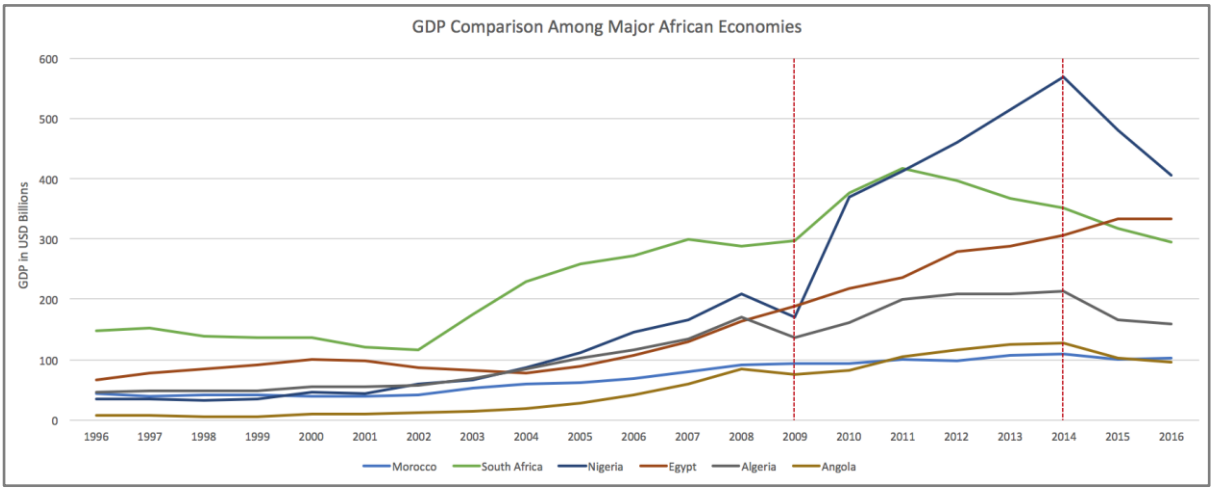


Figure 1: GDP Comparison Among Major African Economies

2.2 ECI Ranking and Changes

The Economic Complexity Index (ECI), as defined by Ricardo Hausmann and Cesar Hidalgo in 2009 to explain the knowledge accumulated in a country's population, exhibits similar pattern for Egypt. As illustratedin the Figure 2 below, Egypt exhibits stability in terms of its production capability from 2009 to 2014, as measured by ECI. This is in direct contrast to countries such as Algeria and Morocco, which demonstratea disadvantage as compared withEgypt in terms of both the GDP and the ECI indicators.

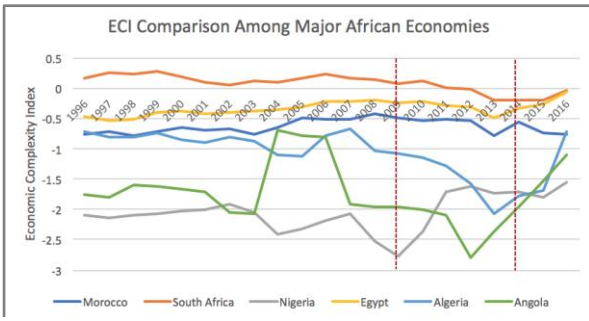


Figure 2: ECI Comparison

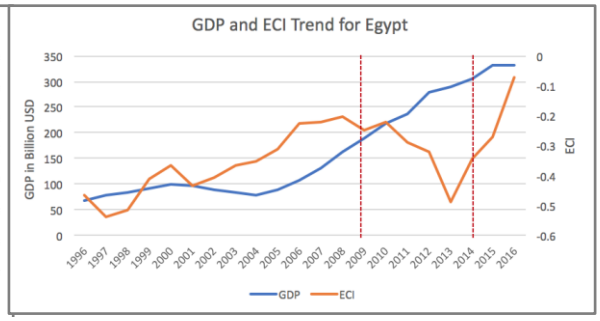


Figure 3: GDP and ECI (Egypt)

Figure 3 offers a further comparison of GDP and ECI over the entire time span from 1996 to 2016 for Egypt. Based on statistical analysis, ECI is a significant covariate for GDP with an adjusted R-square of 17.8%. However, the relationship is reversed during the period of our interest and we observe a general decrease in ECI during the period. The results show that the economic growth experienced by Egypt during the period from 2009 to 2014, though stable, is not very consistent with the predictions by ECI and Egypt had not yet effectively developed a competitive product space that helps to accumulate complicated knowledge and technology. In order to provide a more concrete answer to why this has happened, we will dive into an analysis of Egypt's production structure and its product space designs in the next few chapters.

3. ProductionAnalysis

3.1 Product Categorization

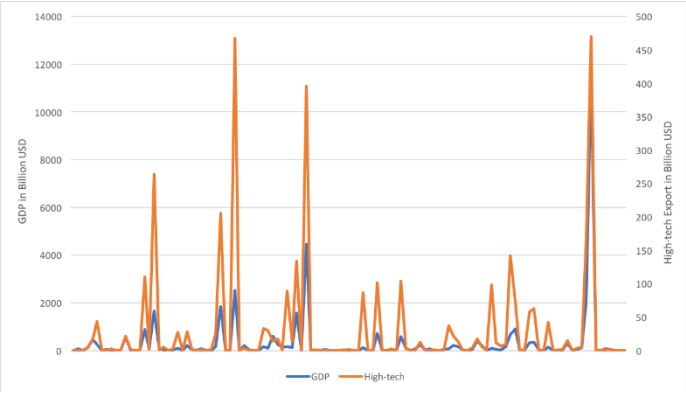
Our main data source in this research is the UN Comtrade Website (https://comtrade.un.org), which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to have a holistic vision of the product space of a country, however, a level of categorization is required. Product in our data source are labelled from 010000-999999, with the first two-digits representing a general category to which the product belongs. Based on the 99 categories as extracted from the label, we further group the products into 18 most widely-accepted industries, as summarized in Table 1.

Note that the high-tech industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technical gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), high-tech industry has the highest PfCI among all industry groups.

Category	Description
Animal and By-product	All kinds of meat, livestock and related by-products
Vegetable, Fruit and Plant	All kinds of plants, edible vegetables and fruits
Food and Preparation	Edible preparations, raw cooking materials and appliances
Mineral and Stone	Mining industry for mineral and stone materials
Chemical	Raw chemical materials and by-products
Chemical Manufacturing	Advanced chemical products and pharmaceutical products
Raw Manufacturing	Basic products made from plastic, rubber, leather, etc.
Wood Product	Book, paper and other elementary products related to wood
Clothing and Accessory	All kinds of fabrics, filament, clothing and accessories
Textile	All kinds of textile articles and related products
Metal	Mining of all sorts of raw metal
Machinery	Machinery and mechanical appliances, including electrical equipment
Transportation	Vehicles and locomotives including aircraft, ship, railway, etc.
Weapon	Explosives, arms and ammunition
Daily Instrument	Basic daily instruments such as sport requisites, timing and measuring tools
Precious Stone	Refined or processed stone products such as pearl and ceramics
Service	Financial, business, legal, communication, travel services, etc.
High-tech	Agglomeration of advanced products with highest PCIs

Table 1: Categorization of Products

We further observe a strong positivecorrelation between the GDP and the export of high-tech products. As shown by Figure 4 below, a high adjusted R-square of 67.6% exists between GDP and high-tech exports, indicating that more successful economics, as measure by GDP, tend to be those that are capable of producing and exporting high-tech products as we define them.



Statistics	Value
Multiple R	0.824
R-Square	0.679
Adjusted R^2	0.676
Variable	**Value**
Coefficient	12.20
t-stat	15.61
P-value	0(***)

Figure 4: Regression of GDP and High-tech Exports

3.2 Actual Development

Based on the product categorization aforementioned, we aim to capture Egypt's actual production strategy between 2009 and 2014 on a complexity level.In Hausmann and Hidalgo (2009), distance is defined to be a measure of how far a country is from a certain product not in its product space. We extend this concept to all product groups and quantify how technically equipped a country is to develop corresponding product groups. In other words, the shorter the distance between a country and a product group, the more capable a country is in developing products within that group. Defining this net distance change as the Actual Developmental Index (ADI), our export data allows us to calculate and compare ADIofEgypt

and the 18 product groups in 2009 and 2014 through subtraction, where a positive ADI indicates a shortening of the distance (i.e., advancement of the industry), and a negative ADI indicates the reverse. As ranked in Table 2, during the period from 2009to 2014, Egypt has focused mostly on products of low complexity and those requiring basic skills, including those related to animal and vegetable.

Category	ADI	Category	ADI
1. Animal and By-product	0.324	2. Service	0.298
3. Weapon	0.296	4. Vegetable, Fruit and Plant	0.294
5. Mineral and Stone	0.292	6. Wood Product	0.291
7. Clothing and Accessory	0.291	8. Textile	0.270
9. Raw Manufacturing	0.261	10. Chemicals	0.247
11. Daily Instrument	0.239	12. Precious Stone	0.224
13. Chemical Manufacturing	0.211	14. Transportation	0.210
15. Food and Preparation	0.201	16. Metal	0.200
17. High-tech	0.135	18. Machinery	0.132

Table 2: Actual Developmental Index (ADI) of Egypt

3.3 Regional and Global Comparison

Table 3and 4 below summarizes how the production strategy adopted by Egypt compares with its African counterparts and a selected group of leading economies in the world, as measured in ADI. It can be directly observed that Egypt focused exclusively on products with low complexity level,as is common among its African counterparts such as South Africa and Morocco, which give a considerable level of attention to basic product groups such as animal and by product, precious stone and wood product.

Ranking	Egypt	South Africa	Morocco
1	Animal and By-product	Animal and By-product	Chemical Manufacturing
2	Service	Raw Manufacturing	Precious Stone
3	Weapon	Daily Instrument	Wood Product
4	Vegetable, Fruit and Plant	Food and Preparation	Textile
5	Mineral and Stone	Vegetable, Fruit and Plant	Clothing and Accessory

Table 3: ADIComparison on a Regional Scale

Ranking	USA	Germany	Singapore	China
1	Service	Chemicals	Machinery	Precious Stone
2	Chemical Manufacturing	Service	Transportation	Service
3	Food and Preparation	High-tech	Daily Instrument	High-tech
4	Textile	Transportation	Chemicals	Machinery
5	Clothing and Accessory	Food and Prep	High-tech	Chemicals

Table 4: ADI Comparison on a Global Scale

Table 4 further examines Egypt’s development strategy against those of more developed economies on a global scale. Reasonably, developed economics such as USA, Germany and Singapore have devoted the majority of their efforts into highly complicated industry groups while economies such as Egypt still partially relies on basic industry groups like animal and by-product. This trend holds true in general, regardless of the continent or the time period, and serves as an indicator for which phase of development a country is currently in as

evaluated from their production strategy. Based on this standard, Egypt is indeed adopting an unambitious and moderate strategy that seeks development in its basic industry groups.

4. Product Space Design

4.1 Suggested Development Vector

As aforementioned, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined inHausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP=0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter in the product as compared to global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in term of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP=1) has an opportunity gain of zero. And unreasonable investment on certain product p would lead to a relative decrement in export of more profitable products, and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economics strength, those having higher ECI, or in other words, more reasonable economics structure isoften in better shape.

Ranking	Egypt	South Africa	Morocco
1	Animal and By-product	Clothing and Accessory	High-tech
2	High-tech	Service	Machinery
3	Machinery	High-tech	Daily Instrument
4	Daily Instrument	Daily Instrument	Transportation
5	Transportation	Machinery	Textile
6	Weapon	Textile	Chemical Manufacturing
7	Chemicals	Chemical Manufacturing	Raw Manufacturing
8	Raw Manufacturing	Raw Manufacturing	Weapon
9	Vegetable, Fruit and Plant	Animal and By-product	Wood Product
10	Food and Preparation	Transportation	Precious Stone
11	Mineral and Stone	Weapon	Metal
12	Chemical Manufacturing	Chemicals	Food and Preparation
13	Wood Product	Vegetable, Fruit and Plant	Mineral and Stone
14	Clothing and Accessory	Food and Preparation	Clothing and Accessory
15	Textile	Mineral and Stone	Service
16	Metal	Wood Product	Animal and By-product
17	Precious Stone	Metal	Chemicals
18	Service	Precious Stone	Vegetable, Fruit and Plant
ECI	-0.246	0.074	-0.495

Table 5: Opportunity Gain Ranking on a Regional Scale

We define the ranking of opportunity gain for a country c, as the Suggested Development Vector (SDV) of c. Table 5 and 6 summarize how the SDV ofEgypt compares with a selected group of regional economies and leading economies in the world, with the industry groups ranked in decreasing order of significance. Dark grey indicates positive opportunity gain, gray indicates zero opportunity gains, and light grey indicates negative opportunity gains. The result shows that leading countries barely have any industries with positive opportunity gain. In other words, those countries are already in the most stable and efficient economic structure. Without the necessity of economic transition, those countries merely need to maintain a steady and balanced economic development. In contrast, developing countries have nearly half of industries with positive opportunity gain, and share common interests in industrial manufacturing and high-tech industries. It can be concluded that Egypt shares the

characters of developing countries, with a premature economics structure that leans towards basic to intermediate levelindustries.

Ranking	USA	Germany	Singapore	China
1	Machinery	Machinery	Machinery	Machinery
2	High-tech	High-tech	High-tech	High-tech
3	Transportation	Transportation	Chemicals	Chemicals
4	Chemicals	Metal	Mineral and Stone	Daily Instrument
5	Textile	Raw Manufacturing	Transportation	Textile
6	Clothing and Accessory	Daily Instrument	Weapon	Metal
7	Metal	Chemical Manufacturing	Chemical Manufacturing	Raw Manufacturing
8	Raw Manufacturing	Wood Product	Daily Instrument	Wood Product
9	Daily Instrument	Weapon	Textile	Clothing and Accessory
10	Chemical Manufacturing	Service	Metal	Service
11	Weapon	Chemicals	Raw Manufacturing	Chemical Manufacturing
12	Wood Product	Clothing and Accessory	Precious Stone	Precious Stone
13	Service	Mineral and Stone	Wood Product	Weapon
14	Vegetable, Fruit and Plant	Animal and By-product	Clothing and Accessory	Animal and By-product
15	Precious Stone	Precious Stone	Animal and By-product	Mineral and Stone
16	Animal and By-product	Textile	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Food and Preparation	Food and Preparation	Service	Food and Preparation
18	Mineral and Stone	Vegetable, Fruit and Plant	Food and Preparation	Transportation
ECI	1.533	1.917	1.578	0.754

Table 6: Opportunity Gain Ranking on a Global Scale

4.2 Sequential Development Strategy

The sequential economic development strategy aims to approximate the most efficient order of industry development.As discussed in the previous section, opportunity gain measures the potential profit in terms of economic complexity when a country invests on a product p to an extent such that the MCP bumps up from 0 to 1 (i.e., the country becomes a significant exporter in that product). Obviously, the difficulty of bumping up MCP varies across industries. Therefore, we capture this variation of difficulty by defining the concept of marginal opportunity gains.

$$marginal\ opportunity\ gains_{cp} = \frac{opportunity\ gains_{cp}}{\left(\frac{minimum\ \ increment\ \ of\ export\ \ of\ product\ \ p\ for\ country\ \ c}{present\ \ export\ \ of\ product\ \ p\ for\ country\ \ c}\right)}$$

In each iteration, we order all 18 industries based on their current marginal opportunity gain, and prioritize the industry with the highest marginal opportunity gain. We increase the prioritized industry's export to an extent that the MCP firmly bumps up to 1.Following the routine above, we derive the iterative strategy as shown in Table 6, ranked by marginal opportunity gains in decreasing order. Dark grey indicates positive opportunity gain, gray indicates zero opportunity gains, and light grey indicates negative opportunity gains.

4.3Strategy Interpretation

Table 7 shows that at present Egypt lacks necessary industrial power, and therefore barely has advantage in manufactured goods. Basic products such as animal and by-product, wood product are the foundation of intermediate-level products, and industries such as raw manufacturing, daily instrument and transportation are the second easiest step. As a result, gaining comparative edges in these industries will rocket the ECI of Egypt, and it is noteworthy that the significance of high-tech drops extensively after the development of servicere-opens the door of light industries and intermediate-level industries. Because Egypt has a large gap between its current product space and advanced product groups such as high-tech and machinery, it is extremely costly for Egypt to develop these two products even in the

final stable stage, and Egypt may have a compelling industrial structure without a comparative edge in these two industry groups.

Before Iteration	Iteration 1	Iteration 2	Iteration 3
Weapon	Animal and By-product	Chemicals	Wood Product
Animal and By-product	Chemicals	Raw Manufacturing	Raw Manufacturing
Chemicals	Raw Manufacturing	Daily Instrument	Daily Instrument
Raw Manufacturing	Daily Instrument	Transportation	Transportation
Daily Instrument	Transportation	Machinery	Machinery
Transportation	Machinery	High-tech	High-tech
Machinery	High-tech	Animal and By-product	Animal and By-product
High-tech	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
Vegetable, Fruit and Plant	Food and Preparation	Food and Preparation	Food and Preparation
Food and Preparation	Mineral and Stone	Mineral and Stone	Mineral and Stone
Mineral and Stone	Chemical Manufacturing	Chemical Manufacturing	Chemicals
Chemical Manufacturing	Wood Product	Wood Product	Chemical Manufacturing
Wood Product	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
Clothing and Accessory	Textile	Textile	Textile
Textile	Metal	Metal	Metal
Metal	Weapon	Weapon	Weapon
Precious Stone	Precious Stone	Precious Stone	Precious Stone
Service	Service	Service	Service

Iteration 4	Iteration 5	Iteration 6	Iteration 7
Raw Manufacturing	Daily Instrument	Transportation	Animal and By-product
Daily Instrument	Transportation	Animal and By-product	Vegetable, Fruit and Plant
Transportation	Machinery	Vegetable, Fruit and Plant	Food and Preparation
Machinery	High-tech	Food and Preparation	Mineral and Stone
High-tech	Animal and By-product	Mineral and Stone	Chemicals
Animal and By-product	Vegetable, Fruit and Plant	Chemicals	Chemical Manufacturing
Vegetable, Fruit and Plant	Food and Preparation	Chemical Manufacturing	Raw Manufacturing
Food and Preparation	Mineral and Stone	Raw Manufacturing	Wood Product
Mineral and Stone	Chemicals	Wood Product	Clothing and Accessory
Chemicals	Chemical Manufacturing	Clothing and Accessory	Textile
Chemical Manufacturing	Raw Manufacturing	Textile	Metal
Wood Product	Wood Product	Metal	Transportation
Clothing and Accessory	Clothing and Accessory	Weapon	Weapon
Textile	Textile	Daily Instrument	Daily Instrument
Metal	Metal	Precious Stone	Precious Stone
Weapon	Weapon	Service	Service
Precious Stone	Precious Stone	High-tech	High-tech
Service	Service	Machinery	Machinery

Table 7: Marginal Opportunity Gain in an Iterative Process

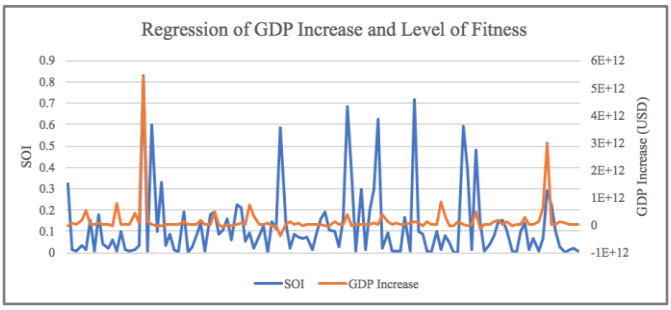
Note that high-tech and machinery show negative marginal opportunity gain at the final stable stage, but that does not mean the two industries are valueless for the development Egypt. Instead, the suggestion means that Egypt, given to its economic situation at the final stage, is encouraged to prioritize the other industries and develop those bottom two industries only after it builds a balanced economic structure in the first place. The industries with negative opportunity gains, though not prioritized in their particular stage, should therefore still be proportionately strengthened within the capability of the country and not left behind.

5. Macro-level Predictions

5.1 Structural Optimality Index

Indeed, such choice of prioritization has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how effective our developmental strategies are and how deviations from the suggested developmental path will lead to non-optimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the R-Square obtained from regressing the country's actual development over the suggested development strategy. Both strategies are quantified in nature, with the actual development represented by export increase in major industry groups and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The R-square from the regression thus provides us with a concrete measure of how the development path adopted by the country has fitted with/differed from our suggestion.

After obtaining this measure for all countries in our database, we further regress countries' five-year economic growth, which as defined as the net increase in GDP (USD), over their SOI. Figure 6 illustrates the regression output, which shows a strong positive correlation between countries' level of fitness and their five-year GDP growth, significant at the 1% level and achieving an adjusted R^2 of 11.9%. In other words, the more a country develops as suggested by calculations on a complexity level, the better economic outcomes it achieves.



Statistics	Value
Multiple R	0.355
R-Square	0.126
Adjusted R^2	0.119
Variable	**Value**
Coefficient	$1.249*10^{12}$
t-stat	4.061
P-value	0.000(***)

Figure 6: Regression of Five-year GDP Growth over SOI

5.2 SOI of Egypt and Its Implications

Egypt, for instance, has a moderate level of structural optimality as compared with its emerging peers. As the regression in Figure 7 suggests (the orange curve represents export increase and the blue curve stands for opportunity gain of the corresponding industry), Egypt demonstrates an average level of correlation with the suggested developmental path, having an adjusted R-square of around 19.2%. Egypt ranks the 22^{ed} place among all countries in terms of SOI and has made an average rate of economic progress among its emerging peers from 2009 to 2014, which is consistent with the regression result in Figure 6. This is in direct contrast to emerging economies such as China, where there is high SOI of around 82.8% and a corresponding GDP growth of 5.39 billion USD, making it one of the fastest developing economies during the period. In sum, Egypt'sstable economic progress from 2009 to 2014 can be at least partially explained by a development strategy that demonstrates a considerable level of matcheswith our suggested developmental path. In specific, Egypt invested heavily in the service industry, while it should have focused more on moderately complex industry groups such as chemical manufacturing.

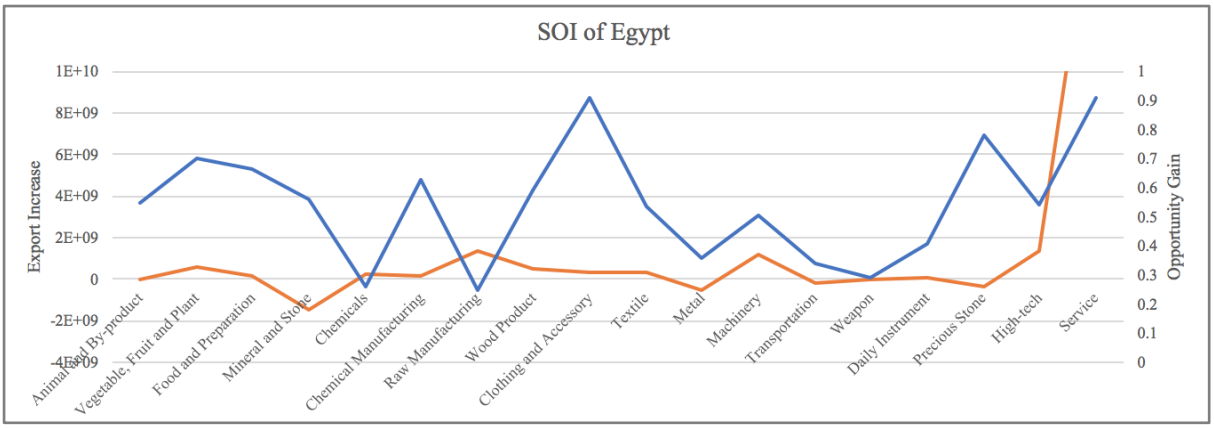


Figure 7: SOI ofEgypt (2009-2014)

6. Concluding Remarks

In conclusion, Egypt has demonstrated a moderate level of economic growth in terms of GDP during the period from 2009 to 2014, concurrent with a stagnation in economic complexity as compared with other major African economies.

The level of economic progress during the period, we believe, can be partially explained through analysis of the economic and product complexity structure of Egypt on the industry group level. The product classification system we adopt allow us to measure the actual development for various industry groups and to observe that Egypt invested heavily in developing animal, vegetable, mineral and stone industries during the period, while its emerging peers and developed economies around the globe have adopted much more complex and progressive paths.

Based on our analysis of the opportunity gain and a sensitivity analysis of major industry groups, Egypt could further its gains by high-tech, machinery, and daily instrument. We further note that Egypt has a slightly matched development strategy that indicates a moderate level of structural optimality on the complexity level, having an adjusted R-square of around 19.2%. Since a significant positive correlation exists between five-year GDP growth and the structural optimality index of an economy as we define and calculate, Egypt indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale, and has the potential to further its economic gains through industrial structure optimization as suggested in the paper.